Leveraging Vision Foundation Models for Semantic Implicit Multi-Sensor SLAM

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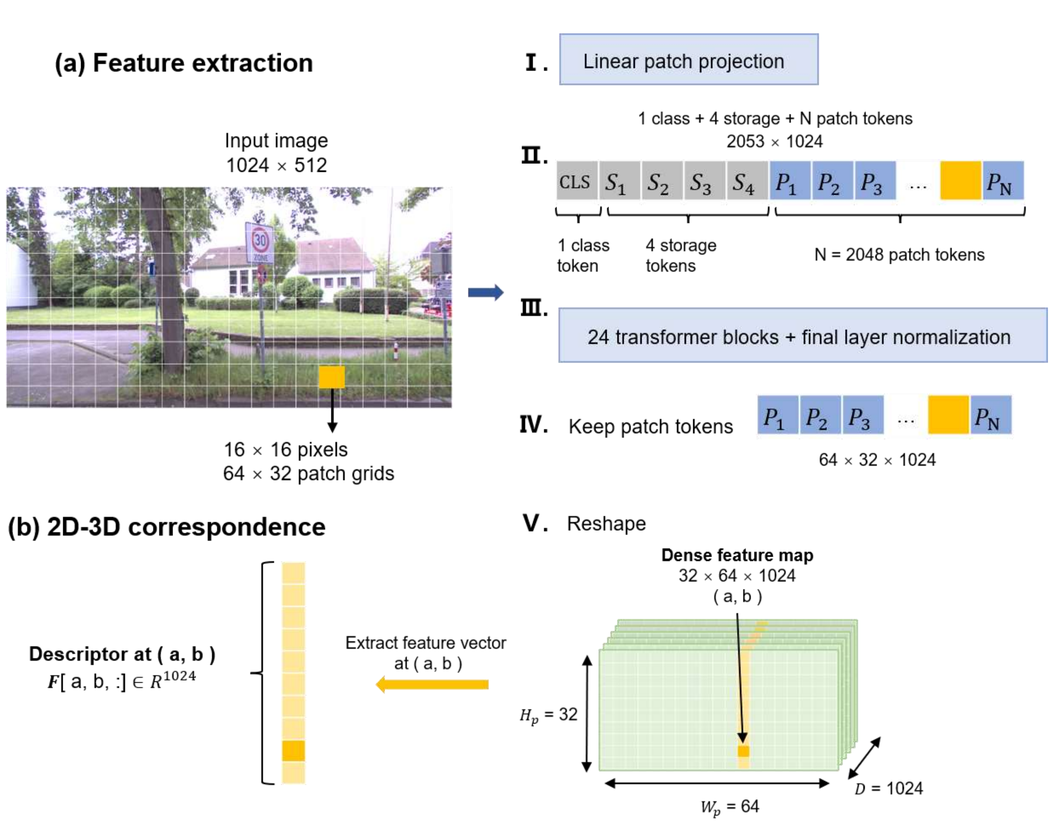
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Abstract

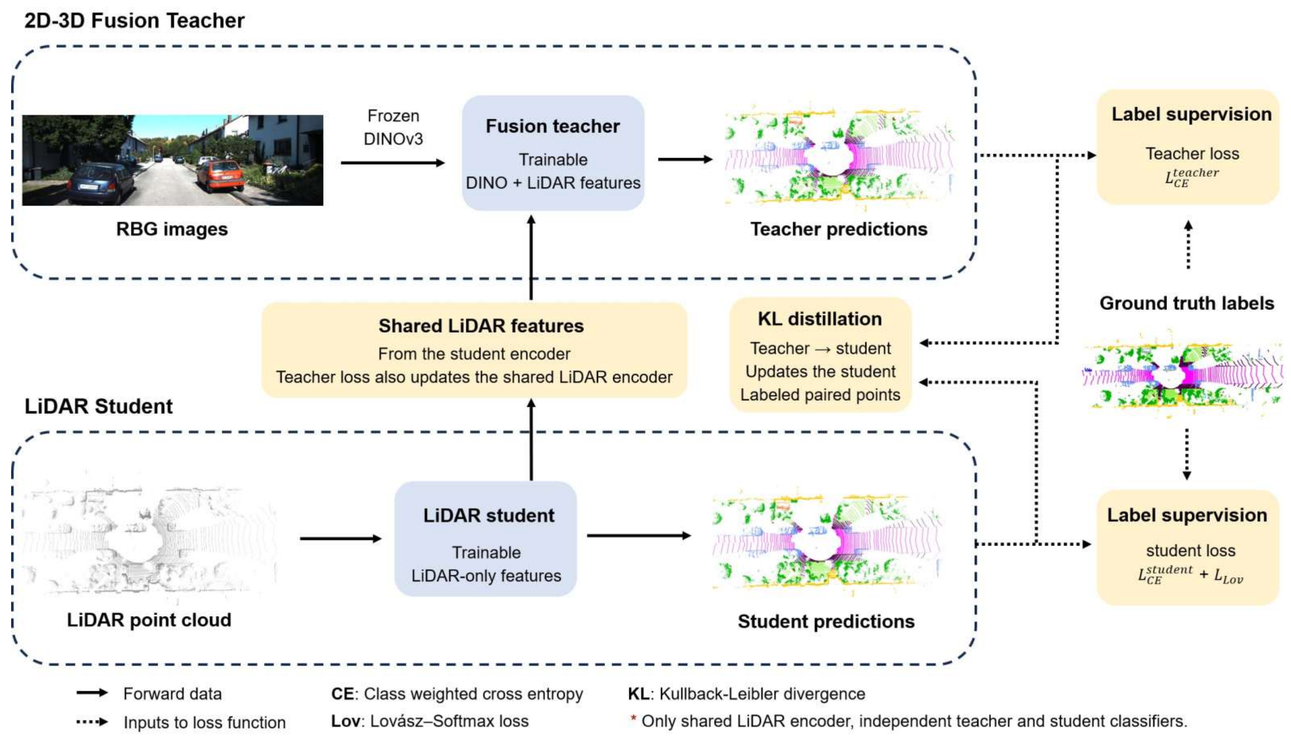
- Semantic SLAM, Semantic segmentation
- Why? Semantic information supports scene understanding and interaction with the environment.
- How? We jointly train a 2D-3D fusion teacher and a LiDAR-only student using frozen DINOv3 features and multi-scale fusion-to-single knowledge distillation.
- Results? The student achieves 63.5% SemanticKITTI test mIoU at 16.5 FPS for segmentation inference. Semantic map supervision also reduces aligned APE.

Method

- Calibrated 2D-3D correspondence associates LiDAR points with DINOv3 2D features
- Based on a multi-scale fusion-to-single knowledge distillation, 2D-3D fusion teacher and LiDAR-only student model share a LiDAR encoder and train jointly with supervised segmentation.



Schematic of patch feature extraction with frozen DINOv3 ViT-L/16. The final patch descriptors retain their spatial indices and form the grid used for LiDAR–image association.



Overview of pipeline for LiDAR semantic segmentation. The fusion teacher combines image features from a frozen DINOv3 encoder with LiDAR features from the student encoder. Ground truth point labels supervise teacher and student model, while KL distillation aligns the student's class probabilities with those of the teacher on image paired points with valid labels. Teacher and student model share the LiDAR encoder and use separate classifiers. Only the LiDAR student is used at inference.

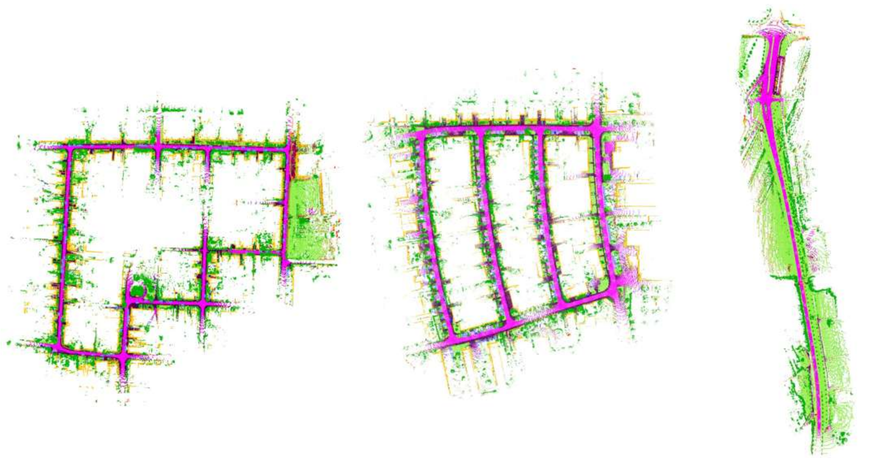
Experiments and Results

- It is capable of effectively using vision foundation model for semantic segmentation.
- We also showcase the ability of our approach to work effectively within a SLAM pipeline for building semantic mapping.
- Our method achieves 63.5% mIoU on official SemanticKITTI single-scan sequence 11-21.

REPRESENTATIVE SEMANTICKITTI TEST RESULTS

Method	Inference input	mIoU (%)
KPConv [23]	L	58.8
FusionNet [24]	L	61.3
KPRNet [25]	L	63.1
Lite-HDSeg [26]	L	63.8
2DPASS [3]	L	72.9
DITR [13]	L+C	74.4
LSK3DNet [9]	L	75.6
Baseline $+W$ (ours)	L	63.5

Single-scan test sequences 11–21. L: LiDAR; C: RGB images. Input denotes the evaluated inference configuration. The DITR entry uses image-feature injection.



Prediction results on SemanticKITTI sequence 15-17



Semantic mapping on IPB dataset